

AWR2944EVM DSS Program Manual(Windows 11)

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Requirements:

Hardware:

- AWR2944EVM Board
- 3.5mm power plug with 12V and less than 2.5A output
- USB-A to Micro-USB Cable

Software:

- [Code Composer Studio 20](#)
- [UNIFLASH](#)

Github File(Request Access):

1. Go to [DSS Radar Program – Standalone](#) page
2. Navigate to Code<> → Download Zip
 - a. (Alternate) If you have Github Desktop, Copy Repository
3. Extract files to a chosen folder.

For ease of functionality, install CCS 20 and Uniflash under same ti folder for consistency, prefer directly in your root folder (eg. C:/ti)

This way during operation of the CCS, the different SDK and Example codes can more consistently refer to the correct file-paths.

For purposes of this tutorial, the repo code is installed in C:\Git\Radar_software, and will be referred to during the tutorial. The final code location depends on where you install it.

Terminology

SDK – Software Development Kit

OOB – Out of Box

TI – Texas Instruments

CCS – Code Composer Studio

Introduction

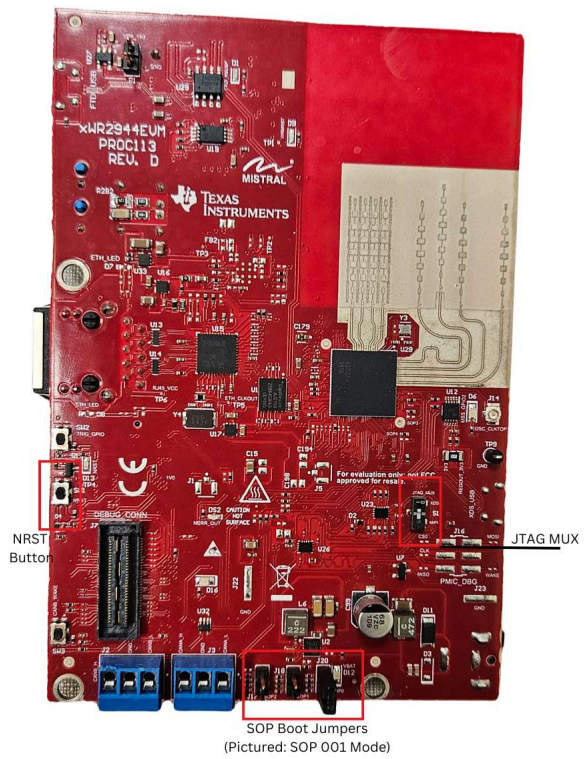
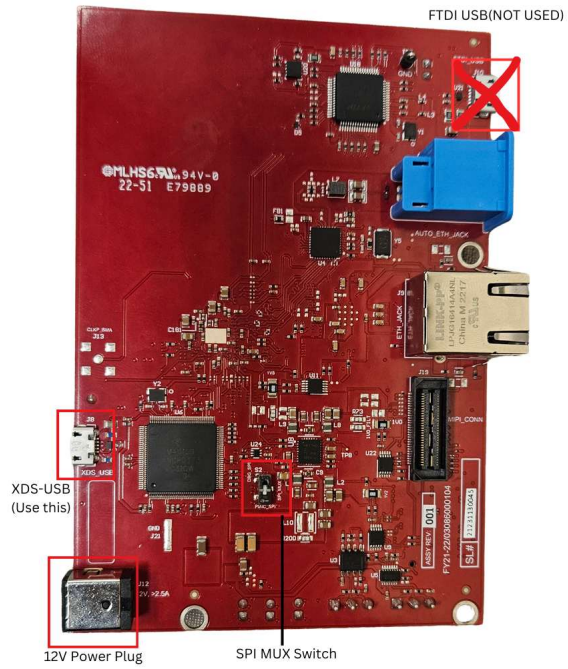
This is an alternate manual for setting up the pre-edited Distributed Sensor Systems version of the original Texas Instruments Out of Box Demo. If you are looking for the guide to set up the un-edited Out of Box Demo, refer to the AWR2944EVM Out of Box Demo Manual, instead.

This manual provides the steps for accessing the pre-made, edited and simplified version of the Out of Box demo created as part of the 2026 Third Year internship semester. This code should give a more straight-forward point of entry for programming the AWR2944EVM radar board without as much of the set-up asked for by the original Out of Box demo.

Much of this manual will repeat steps of the Out of Box manual, with main differences beginning at Step 6.

Out of Box Default	DSS Radar Program
Doppler Division Multiplex	Time-Division Multiplexing
Ethernet Incompatible	Ethernet Compatible
	Memory Linker Optimization
	Simplified Main Code File
	Closer to final implementation design

The AWR2944EVM Board is notably temperamental where not following the exact order of operations can cause the board to not function as expected, therefore it is important to carefully read the manual first, then follow along the instructions. It is possible some steps are more optional, but it is good to first follow the examples detailed here in order to know the steps currently known to work.

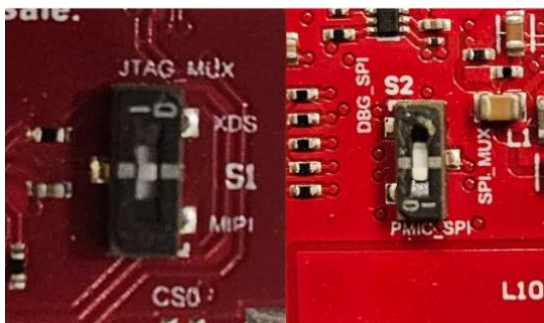


Hardware + Firmware Setup

The AWR2944EVM Radar Board enables different boot modes that allow user to flash firmware or set it to a functional mode that runs the flashed firmware. In order to set the board so that it can use and simultaneously edit the OOB source code, it needs to be configured with specific debug firmware.

Preparation:

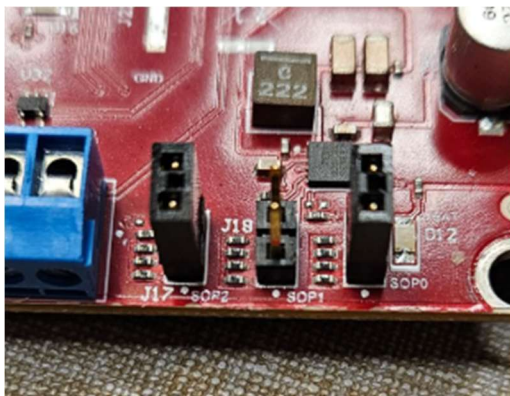
Set the XPI and JTAG Mux Switches as follows:



JTAG MUX: None(Middle)

SPI_MUX : PMIC_SPI

Step 1: Set the SOP Pins to 101 Mode (Flash)



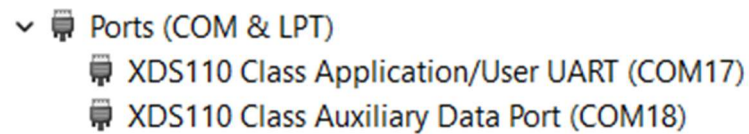
The SOP Configuration equals to which headers are “closed” by a jumper. 1 = Jumper/Closed
0 = No Jumper / Open, therefore the pictured configuration equals to SOP 101, used for flashing the firmware.

Step 2: Connecting the Radar Board

1. Power the radar board by connecting the 12V power jack.
2. Press the NRST Button (Required for the board to boot correctly)
3. Connect Board to computer by connecting the Micro-USB cable to the XDS-USB Port(Located next to the power jack port)

Step 3: Confirm COM Port Connection

1. Open computer Device Manager
2. Navigate to Ports (COM & LPT)
3. Confirm the following ports are displayed:
 - a. XDS110 Class Application/User UART
 - b. XDS110 Class Auxiliary Data Port

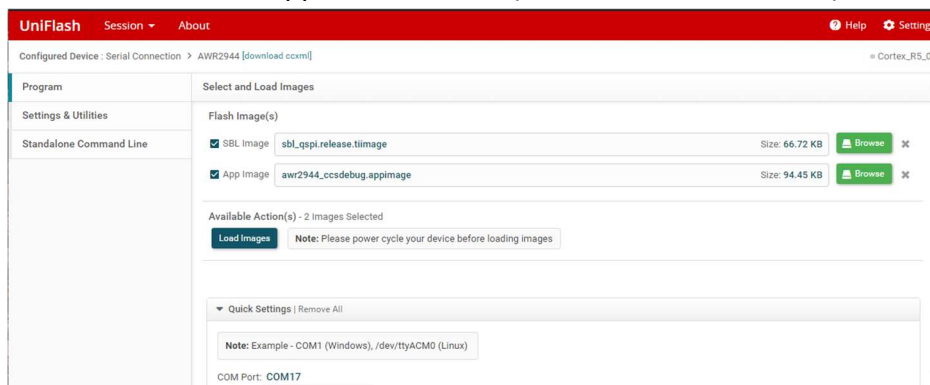


Example view of the correct COM Ports (COM number values will differ by system)

If the com ports are not displayed correctly or have a yellow warning icon, it could be the system does not have the correct drivers installed, in which case the drivers can be installed [here](#).

Step 4: Flash Firmware using Uniflash

1. Start Uniflash
2. AWR2944EVM should show up in the Detected Devices on the top of the window.
 - a. Otherwise find AWR2944EVM in the New Configuration Menu below
3. Press Start to begin configuring the board.
4. The next window has following sections of importance:
 - a. SBL Image
 - b. App Image
 - c. Quick Settings: COM Port
5. For SBL Image:
 - a. Click Browse
 - b. Navigate to:
C:\Git\Radare_software\Dependencies\mmwave_mcuplus_sdk_04_07_02_01\tools\awr294x
 - c. Select sbl_qspi.release.tiimage
6. For App Image:
 - a. Click Browse
 - b. Navigate to:
C:\Git\Radare_software\Dependencies\mmwave_mcuplus_sdk_04_07_02_01\tools\utils\ccsdebug
 - c. Select awr2944_ccsdebug.appimage
7. For Com Port
 - a. Refer to the Device Manager from Step 3 of this manual
 - b. Check the COM Port for XDS110 Class Application/User UART(In the example above, it is COM17)
 - c. Enter the value to the COM Port Tab
8. The window should appear as follows. (With relevant COM Port)



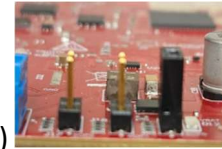
9. After confirming the settings, press Load Image.
10. Console below should produce the following: [SUCCESS] Program Load completed successfully.
11. You can now unplug the power and USB cable from the board for the next step.

Step 5: Importing and Building the Radar Program

1. (If not already) Open CCS
2. From the toolbar, Navigate to File -> Open Workspace from File
3. In the new window, go to C:\Git\Radar_software
4. Select WorkSpace.theia-workspace
5. The file explorer on the left will have Radar Program project with
 - a. Radar_program_dss
 - b. Radar_Program_mss
6. With the projects present, navigate to Project -> Build All
7. CCS Should complete the build for all the projects.
 - a. There will be some errors listed, but these can be ignored for now
8. After the build finishes, confirm it was done correctly by navigating to the code explorer and after expanding any of the projects, expand the "Generated Binaries" tab.
 - a. If there is a file ending in .xe66(in case of dss project) or .xer5f(for mss), the code was built correctly. These are your "applications".
9. This concludes building the initial Radar Program. If the process was successful, it is now possible to edit the code under the dss and mss projects to generate custom applications.

Step 6: Running the Applications on AWR2944EVM

If everything has worked up to this point of the manual, it is now possible to upload the code on the AWR2944EVM board.



1. Change the SOP pins on AWR2944EVM to Functional Mode(001)
2. Power the Radar Board with the 12V connector
3. Press the NRST Button
4. Connect the board to the machine by connecting the Micro-USB cable to XDS-USB port
5. Open CCS with the project files from Step 6
6. In the project explorer navigate to: Radar_Program_dss -> TargetConfigs -> AWR2944.ccxml
7. Right Click on AWR2944.ccxml and select "Start Project-less Debug"
8. After a moment, the CCS should shift to a Debug View with several Tabs on the left-side of the window.
9. (If not already expanded) Expand the "Threads" Tab, here there should be three options.
 - a. Cortex_R5_0 - Disconnected
 - b. Cortex_R5_1 - Disconnected
 - c. C66xx_DSP – Disconnected
10. (Repeat for both Cortex_R5_0 and C66xx_DSP) – Right Click and select Connect Target.
 - a. The status of the core should change from Disconnected to Halted
 - b. Right Click again and go to Reset -> CPU Reset
11. Once you have done this for both "cores"(ignore Cortex_R5_1 for now), click on Cortex_R5_0 to Highlight it.
12. From the top toolbar navigate to: Run - > Load -> Load Program
13. From the new window select "Workspace..."
14. Navigate to: Radar_Program_mss - > Debug -> Radar_Program_mss.xer5f and press Ok, then Ok again.
 - a. The status on Cortex_R5_0 should go from Halted to Halted with a red marker.
15. Repeat previous step on C66xx_DSP but select Radar_Program_dss -> Debug -> Radar_Program_dss.xe66
16. In order to fully run the Program, right click on Cortex_R5_0 and select "Continue", wait a moment and then repeat with C66xx_DSP
17. The program is now running on the AWR2944EVM board. The program is not saved on the board, these step need to be repeated after resetting/powering off the board.

Step 7: Testing the program with the Demo Visualizer

When first setting up the OOB Demo code, it might be desirable to test to see if it works. If the goal is to customize the OOB demo code for a more specific purpose, this step is likely to be irrelevant later on as the Visualizer is likely not to work correctly. But for first time testing, this is still a good step to get familiar with the radar.

Note, these steps should be made directly after point 17 of Step 7.

1. Open the Web-Based Demo Visualizer [Here](#).
 - a. If using the visualizer for the first time, it will prompt installing a plug-in and TI Cloud Agent, follow the instructions on the page to do this.
2. First step is connecting the Hardware. For this step the values of the COM Ports from Step 3 are required, refer to the Device Manager if these have been forgotten.
3. In the Demo Visualizer, select Options -> Serial Port in the Top-Left of the window.
4. There are two Com Port selections that open
 - a. mmWave:User/Application Port (CLI CFG_port)
 - i. For this, enter the COM value that corresponds to XDS110 Class Application/User UART in the Device Manager
 - b. mmWave:Auxillary Data port (Demo output DATA_port)
 - i. For this, enter the COM Value that corresponds to XDS110 Class Auxiliary Data Port in the Device Manager
5. For the Right Side Baud Rate: Open the drop down and scroll to the bottom of the options to find "Custom"
 - a. Enter 3125000 as the value, then press OK.
6. If everything is set up correctly, the Status Bar at the bottom of the window should display "Hardware Connected"
7. Configure The Demo Visualizer window for the following settings:
 - a. Platform: AWR294X : AWR2944
 - b. SDK Version: 4.7
 - c. Antenna Config: Any(For Now)
 - d. Processing Chain : DDM
8. This is all that is required before pressing "Send Config to mmWave Device"
 - a. The console message window rolls through the configuration settings
 - b. After this the "Plots" tab can be clicked on to view the real-time data from the radar.

This is the basic setup for the visualizer, it is encouraged to play around with the settings for the radar and test the different graphs available for use with the demo to test how everything works.

The OOB Demo is limited only to the DDM Mode which will limit the available options.

Editing The Source Code

This project is set up to be a base for development and experimentation of the base Texas Instruments code and editing it locally(or on your own Github branch) is greatly encouraged.

The big difference between this code and the base Out of Box is that the “guts” of the code are separated from the main code file.

For most non-algorithm functions, you will be editing in the `mss_main.c` file. For the Radar Program, this file is greatly simplified to include the Main, the taskscheduler function and the base arrays for the program memory.

The setup code and much of the big functionality of the code is now located in `RadarSetup.c` which includes a lot of the code and set up for the individual tasks of the overall radar code. It is useful to preview this on your own time to get a feel for the different systems involved with the code.

By the time of writing an additional manual is (possibly) included to provide a basic demo for implementing a custom function as part of the Radar Program Code.

Conclusion

This has been the manual for setting up and getting started with the Out of Box code for AWR2944EVM Radar Board. Further revisions will hope to simplify the existing steps, remove any redundant ones and possibly provide extensions to the appendix with more visual information as accessibility to individuals that may suffer difficulties in following heavy text-based manuals.

Appendix

Rev.

02/07/26 – Adjusted to Standalone version of the Github Branch